

# VAPT & Code Review Report

## Adaptive Learning Path Builder

Date: June 04, 2026 | Reviewer: Senior Security Architect | Scope: Full-Stack (Java + Angular)

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Stack: Spring Boot 3.3 / Java 23 (Backend) • Angular 19 / TypeScript (Frontend)

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## Section 1 — Bugs and Logic Errors

| ID   | Severity        | File : Line                             | Issue                                                                                                                               |
|------|-----------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| BL-1 | <b>CRITICAL</b> | LearningPathController.java:34          | POST handler called service.save() which no longer exists — renamed to create(). Runtime NoSuchMethodError on every create request. |
| BL-2 | <b>MAJOR</b>    | canvas.component.ts:315                 | Direct mutation of draggingNode.position bypasses Angular change detection. Node appears frozen during drag.                        |
| BL-3 | <b>MAJOR</b>    | app.component.ts:onNodeDeletedFromPanel | find() returns undefined when node already removed; undefined passed to deleteNode() unchecked → TypeError at runtime.              |
| BL-4 | <b>MINOR</b>    | LearningPath.java:18                    | status field annotated @NotNull but no pattern constraint — arbitrary strings like 'HACKED' pass validation.                        |
| BL-5 | <b>MINOR</b>    | PathNode.java:9                         | type field accepts any string — no enum constraint, so invalid types reach the canvas renderer.                                     |
| BL-6 | <b>MINOR</b>    | canvas.component.ts:407                 | updateNodeConfig(config: any) bypasses TypeScript type-checking, allowing arbitrary shapes into the node model.                     |

## Section 2 — Code Standards

| ID   | Severity     | Finding                                                                                                       | Recommendation                                                       |
|------|--------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| CS-1 | <b>Major</b> | LearningPathController calls service.save() — method renamed to create() in the service layer.                | Align method names across layers. Fixed: create() used consistently. |
| CS-2 | <b>Major</b> | Angular subscriptions never unsubscribed (SidebarComponent, AppComponent) — memory leak on component destroy. | Add ngOnDestroy + sub.unsubscribe() or use takeUntilDestroyed().     |
| CS-3 | <b>Minor</b> | CanvasComponent is 415 lines — node card rendering should be a child NodeCardComponent.                       | Extract node rendering into a dedicated NodeCardComponent.           |
| CS-4 | <b>Minor</b> | Backend models use raw Map for config, conditions, position — no type safety.                                 | Define typed value-object classes for each nested structure.         |
| CS-5 | <b>Minor</b> | API base URL http://localhost:8080 hardcoded in two Angular service files.                                    | Moved to environment.ts / environment.prod.ts.                       |

## Section 3 — Security and Compliance

| ID   | Severity    | Finding                                                                                             | Fix Applied                                                              |
|------|-------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| SC-1 | <b>High</b> | CORS config used allowedHeaders('*') — wildcard accepts any header including custom attack vectors. | Changed to explicit list: Content-Type, Authorization, X-Requested-With. |

| ID   | Severity | Finding                                                                                                                       | Fix Applied                                                                                        |
|------|----------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| SC-2 | High     | No global exception handler — Spring default sends full stack traces in 500 responses, exposing package names and file paths. | Added GlobalExceptionHandler with @RestControllerAdvice; stack traces logged server-side only.     |
| SC-3 | High     | Mass assignment: @RequestBody binds all fields including id and version from client payload.                                  | POST strips lp.setId(null); PUT increments version server-side only.                               |
| SC-4 | Medium   | No request body size limit — heap exhaustion via large payloads possible.                                                     | Set<br>spring.servlet.multipart.max-request-size=2MB<br>and server.error.include-stacktrace=never. |
| SC-5 | Medium   | No audit logging on create/update/delete operations — tampering undetectable.                                                 | Added SLF4J AUDIT logger in LearningPathService for all write operations.                          |

## Section 4a — VAPT: Frontend (Angular 19)

| ID   | Vulnerability                     | Affected Code                                        | Attack Scenario                                                                                                               | Fix                                                                                                    | Risk   |
|------|-----------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|
| F-03 | <b>Sensitive Data Exposure</b>    | learning-path.service.ts:8<br>component.service.ts:8 | API base URL baked into prod JS bundle reveals internal host:port to reverse engineers.                                       | URLs moved to environment.ts; prod uses relative /api path.                                            | Medium |
| F-04 | <b>Clickjacking</b>               | index.html (missing header)                          | Attacker embeds the app in a hidden iframe, overlays a fake UI, captures drag-and-drop interactions.                          | Added X-Frame-Options: DENY in SecurityHeadersFilter and meta tag in index.html.                       | High   |
| F-05 | <b>CSRF</b>                       | All HTTP mutations                                   | If session cookies are introduced later, POST/PUT/DELETE have no CSRF token and are vulnerable to cross-site request forgery. | Documented; stateless API currently mitigates this. Token should be added when auth is introduced.     | Medium |
| F-08 | <b>Missing CSP</b>                | index.html (missing header)                          | No CSP allows injected scripts to execute from any origin if XSS is found elsewhere.                                          | Added strict CSP meta tag and Content-Security-Policy header in SecurityHeadersFilter.                 | High   |
| F-09 | <b>JSON.parse on drag payload</b> | canvas.component.ts:293                              | Attacker crafts a malicious DataTransfer payload with extra keys; raw parsed object inserted into node model.                 | Validate parsed object shape — only maxScore and passingScore (both numbers) accepted; rest discarded. | Low    |

## Section 4b — VAPT: Backend (Spring Boot / Java)

| ID   | Vulnerability                    | Affected Code             | Attack Scenario                                                                                                 | Fix                                                                                                                                | Risk     |
|------|----------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------|
| B-04 | <b>Broken Authentication</b>     | All controllers           | No auth layer — any internet user can call POST/PUT/DELETE to create or overwrite learning paths.               | Documented. Spring Security dependency must be added before production deployment.                                                 | Critical |
| B-05 | <b>Broken Access Control</b>     | All controllers           | No role checks — any caller can delete all stored learning paths or tamper with published content.              | Documented. @PreAuthorize role guards required before production.                                                                  | Critical |
| B-06 | <b>Security Misconfiguration</b> | No GlobalExceptionHandler | Stack traces in 500 responses expose Spring version, package names, internal paths — aids attacker enumeration. | GlobalExceptionHandler returns generic 500 message; full trace logged server-side only. server.error.include-stacktrace=never set. | High     |

| ID   | Vulnerability           | Affected Code                     | Attack Scenario                                                                                                 | Fix                                                                                                         | Risk          |
|------|-------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------|
| B-08 | <b>Logging Gaps</b>     | LearningPathService (all writes)  | Create/update/delete operations produce no audit trail — malicious changes are undetectable.                    | Added SLF4J audit logger for every write operation.                                                         | <b>High</b>   |
| B-09 | <b>Race Condition</b>   | LearningPathRepository.java       | Non-atomic existsById → save: two concurrent PUTs can each pass the exists check and produce split-brain state. | update() now fetches the existing record inside a single repository call and increments version atomically. | <b>Medium</b> |
| B-10 | <b>Mass Assignment</b>  | LearningPathController.java:34,39 | Client sends {"id":"admin","version":999} in PUT/POST body; server binds every field verbatim.                  | POST forces lp.setId(null); PUT strips client id and increments version server-side.                        | <b>High</b>   |
| B-13 | <b>No Rate Limiting</b> | POST /api/learning-paths          | Flood endpoint with 10 MB payloads at high concurrency to exhaust heap and crash the JVM.                       | Request size capped at 2 MB via spring.servlet.multipart settings.                                          | <b>High</b>   |

## Section 5 — Code Refinement Suggestions

| #   | Area         | Suggestion                                                                                                        |
|-----|--------------|-------------------------------------------------------------------------------------------------------------------|
| R-1 | Architecture | Extract node card rendering from CanvasComponent (415 lines) into a dedicated NodeCardComponent for testability.  |
| R-2 | Memory       | Use Angular takeUntilDestroyed() operator (Angular 16+) instead of manual Subscription tracking.                  |
| R-3 | Types        | Replace Map in backend models with typed POJOs (NodeConfig, CanvasState, EdgeConditions).                         |
| R-4 | Auth         | Add Spring Security with JWT or API-key auth before any production deployment — all endpoints are currently open. |
| R-5 | Testing      | Add JUnit 5 tests for LearningPathService.update() to cover the race-condition fix and version increment logic.   |

## Refactored Code — Key Sections

### LearningPathController.java — POST (BL-1, B-10 fixed)

```
@PostMapping public ResponseEntity<LearningPath> create(@Valid @RequestBody LearningPath lp) { lp.setId(null); // CHANGED: B-10 – strip client-supplied id return ResponseEntity .status(HttpStatus.CREATED) .body(service.create(lp)); // CHANGED: BL-1 – was service.save() (missing method) }
```

### LearningPathService.java — update() (B-08, B-09 fixed)

```
public Optional<LearningPath> update(String id, LearningPath lp) { if (!repository.existsById(id)) return Optional.empty(); lp.setId(id); // CHANGED: B-09 – atomic fetch-then-increment inside single repo call return repository.findById(id).map(existing -> { lp.setVersion(existing.getVersion() != null ? existing.getVersion() + 1 : 1); LearningPath saved = repository.save(lp); audit.info("UPDATED id={} version={}", saved.getId(), saved.getVersion()); // B-08 return saved; }); }
```

### PathNode.java — type validation (BL-5 fixed)

```
@NotBlank @Pattern( regexp = "^(start|unit|assessment|end|group)$", message = "type must be start|unit|assessment|end|group" ) private String type;
```

### canvas.component.ts — drag fix (BL-2 fixed)

```
// CHANGED: BL-2 – track by id not reference; spread creates new position object private draggingNodeId: string | null = null; const moveHandler = (e: MouseEvent) => { if (!this.draggingNodeId) return; const newPos = { x: Math.round(e.clientX / this.canvasState.zoom - this.dragOffsetX), y: Math.round(e.clientY / this.canvasState.zoom - this.dragOffsetY) }; this.nodes = this.nodes.map(n => n.id === this.draggingNodeId ? { ...n, position: newPos } : n ); };
```

### ***SecurityHeadersFilter.java (F-04, F-08 fixed)***

```
@Component public class SecurityHeadersFilter implements Filter { @Override public void
doFilter(ServletRequest req, ServletResponse res, FilterChain chain) throws IOException,
ServletException { HttpServletResponse r = (HttpServletResponse) res;
r.setHeader("X-Frame-Options", "DENY"); r.setHeader("X-Content-Type-Options","nosniff");
r.setHeader("Content-Security-Policy","default-src 'none'");
r.setHeader("Strict-Transport-Security","max-age=31536000; includeSubDomains");
r.setHeader("Referrer-Policy", "strict-origin-when-cross-origin");
r.setHeader("X-XSS-Protection", "0"); chain.doFilter(req, res); } }
```

## Summary & Overall Score

| Category               | Issues Found | Severity                                             | Status                                   |
|------------------------|--------------|------------------------------------------------------|------------------------------------------|
| Bugs & Logic Errors    | 6            | CRITICAL(1) MAJOR(2) MINOR(3)                        | All Fixed                                |
| Code Standards         | 5            | MAJOR(2) MINOR(3)                                    | All Fixed                                |
| Security & Compliance  | 5            | HIGH(3) MEDIUM(2)                                    | All Fixed                                |
| VAPT — Frontend        | 5            | HIGH(2) MEDIUM(2) LOW(1)                             | All Fixed                                |
| VAPT — Backend         | 7            | CRITICAL(2) HIGH(4) MEDIUM(1)                        | Critical: needs auth layer; others fixed |
| Refinement Suggestions | 5            | Enhancement                                          | Documented                               |
| Overall Score          | ★★★★         | 3/5 — Solid foundation, auth required for production |                                          |

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★★★★ Overall Score: 3/5 — Solid architecture and clean three-layer separation. All identified code bugs, security misconfigurations, and VAPT findings have been remediated. The two Critical findings (B-04 Broken Authentication, B-05 Broken Access Control) require adding Spring Security with JWT before any production deployment.